

EE4C10

Analog Circuit Design Fundamentals

Week Q1.6

October 10th, 2024

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Track-specific parts

- Options
 - *“What is different at **RF frequencies**?”*
 - Responsible teacher: Prof. Leo de Vreede
 - *“The temperature dependence of CMOS devices for **references and temperature sensors**”*
 - Responsible teacher: Prof. Kofi Makinwa
 - *“**Biomedical CMOS circuits**”*
 - Responsible teachers: Prof. Wouter Serdijn / Dr. Dante Muratore
 - **“Synchronized switching rectifiers for energy harvesting”**
 - Responsible teacher: Dr. Sijun Du, Xinling Yue
- Students must follow one (or more) options
- **Students should deliver one track-specific assignment**

Schedule Week 1.7

Week	Date	Day	Time	Activity	Location	Topic
7	15/Oct	Tue	10:45-12:30	Track-specific lecture (Temp sensors) – Prof. Kofi Makinwa/Dr. Javvaji	AE-Hall F (62.0.15)	Track-specific
7	15/Oct	Tue	10:45-12:30	Track-specific lecture (Bio) – Dr. Dante Muratore	AE-Hall E (62.0.21)	Track-specific
7	17/Oct	Thu	10:45-12:30	Track-specific lecture (energy) – Dr. Sijun Du	ME-Lecture Hall F - Simon Stevin (34.A-0-610)	Track-specific
7	17/Oct	Thu	10:45-12:30	Track-specific lecture (RF) – Prof. Leo de Vreede	ME-Hall J (34.D-1-300)	Track-specific

Deadline for HW delivery: **November 7th**

Today's goal

- Get more familiar with noise analysis and simulation

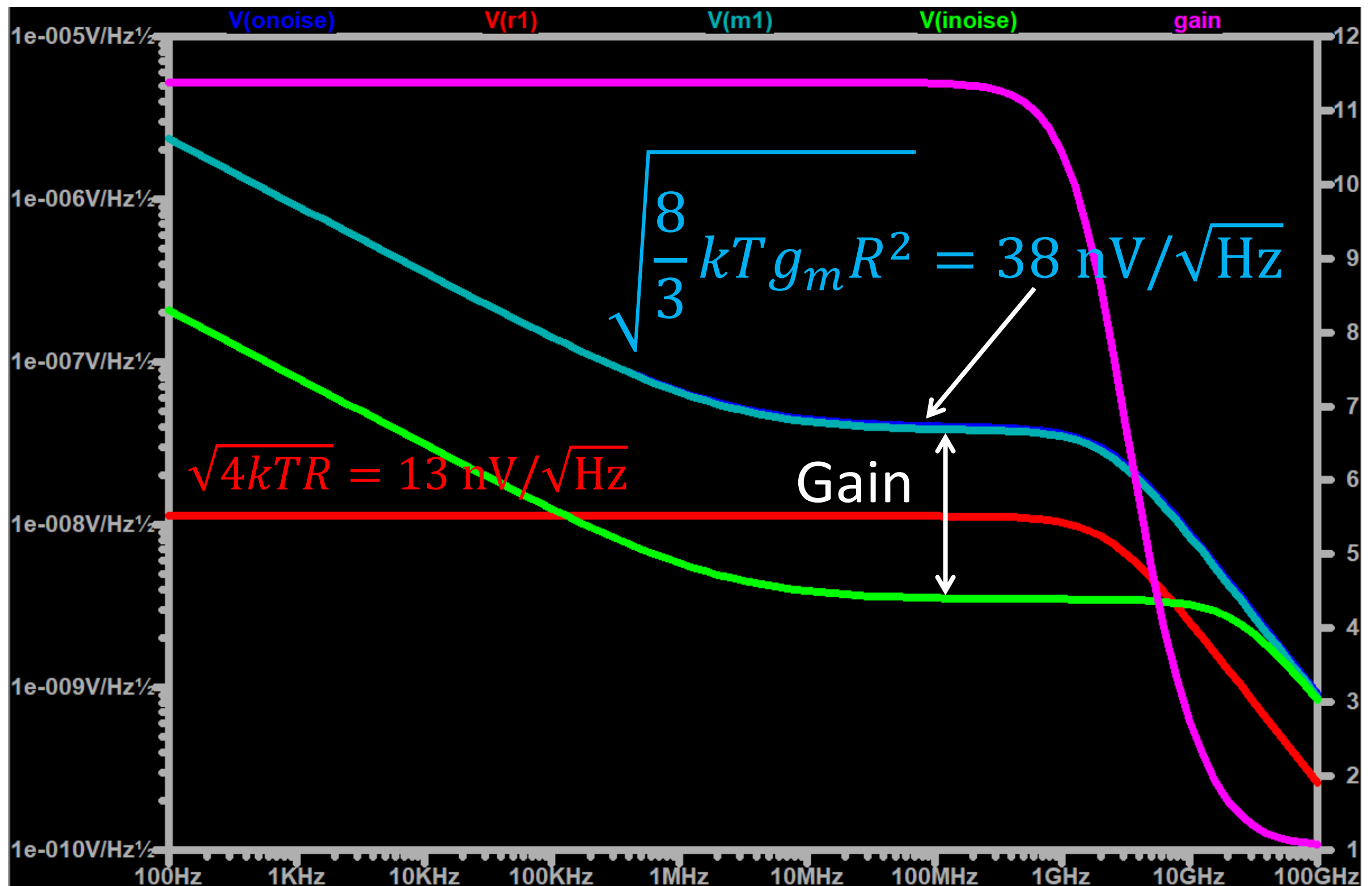
LTspice practical

Exercise 1 – LTspice

Use the file on Brightspace: **Week6_Ex1.asc**. It contains the setup to simulate the noise of resistive-loaded CS amplifier.

- A. Run the noise analysis. Plot the input-referred and output-referred noise.
 - Calculate the noise components using the small-signal models. Use the small-signal parameters from the `.op` analysis (also compare those to your expectations).
 - Compare calculations and simulation. Which noise source is dominating at each frequency range?
- B. Add a cascode with the same size as M1 and bias its gate so that M1 has a margin of 50 mV in VDS from the triode region. To keep both transistors in saturation, reduce the load resistor to 5 k Ω (Why is this required?)
 - Repeat the noise analysis and compare results to your expectations.
 - What changed w.r.t. to simulations in point A?
- C. Reduce the cascode gate voltage by 200 mV and execute the noise analysis again. What changes do you see?
 - Repeat the noise analysis and compare results to your expectations.
 - What changed w.r.t. to simulations in point A?

Results – 1.A

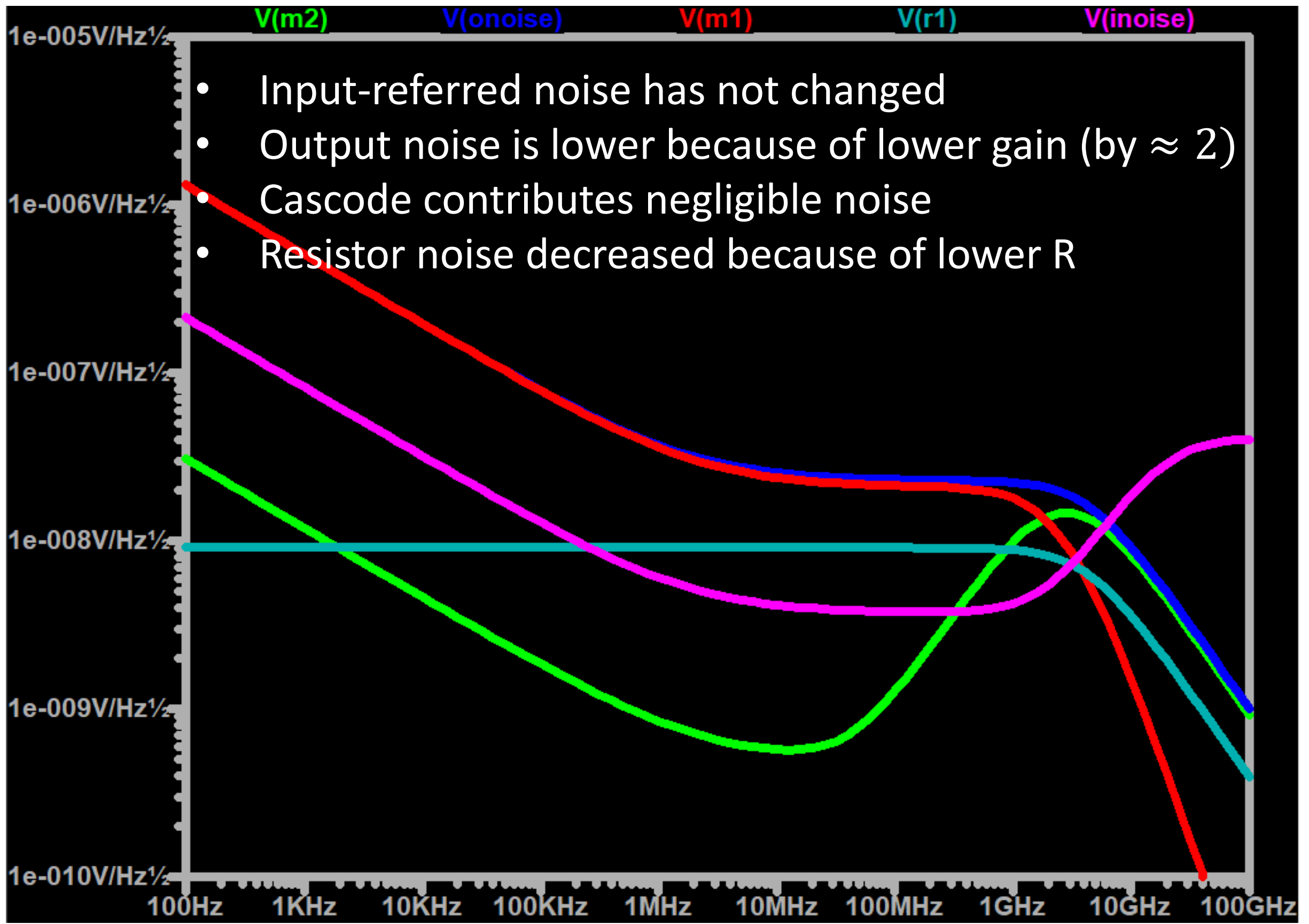


- Note that flicker noise is dominant at low frequencies

Results – 1.B

- A first guess of the cascode gate voltage is
$$V_{G2} = V_{GS1} + (V_{GS1} - V_{th}) + 50 \text{ mV} = 0.9 \text{ V}$$
- However, due to body effect (we assume all NMOS have the bulk to ground), by iteration you can find that it is better to use $V_{G2} \cong 1 \text{ V}$.
- A lower resistance is needed to reduce the drop on the load resistor and have enough headroom for the cascode to be in saturation.

Results – 1.B



Results – 1.C

Main changes

- M1 is now in triode region
 - Gain drops because of lower g_m
- Noise still dominated by M1
 - But different due to different biasing conditions
- Contribution of cascode MOS much higher (order of magnitude)
 - The assumptions needed for neglecting cascode noise is not anymore there
 - Resistance on source of cascode is much lower
 - Due to input transistor in triode

Exercise 2 – LTspice

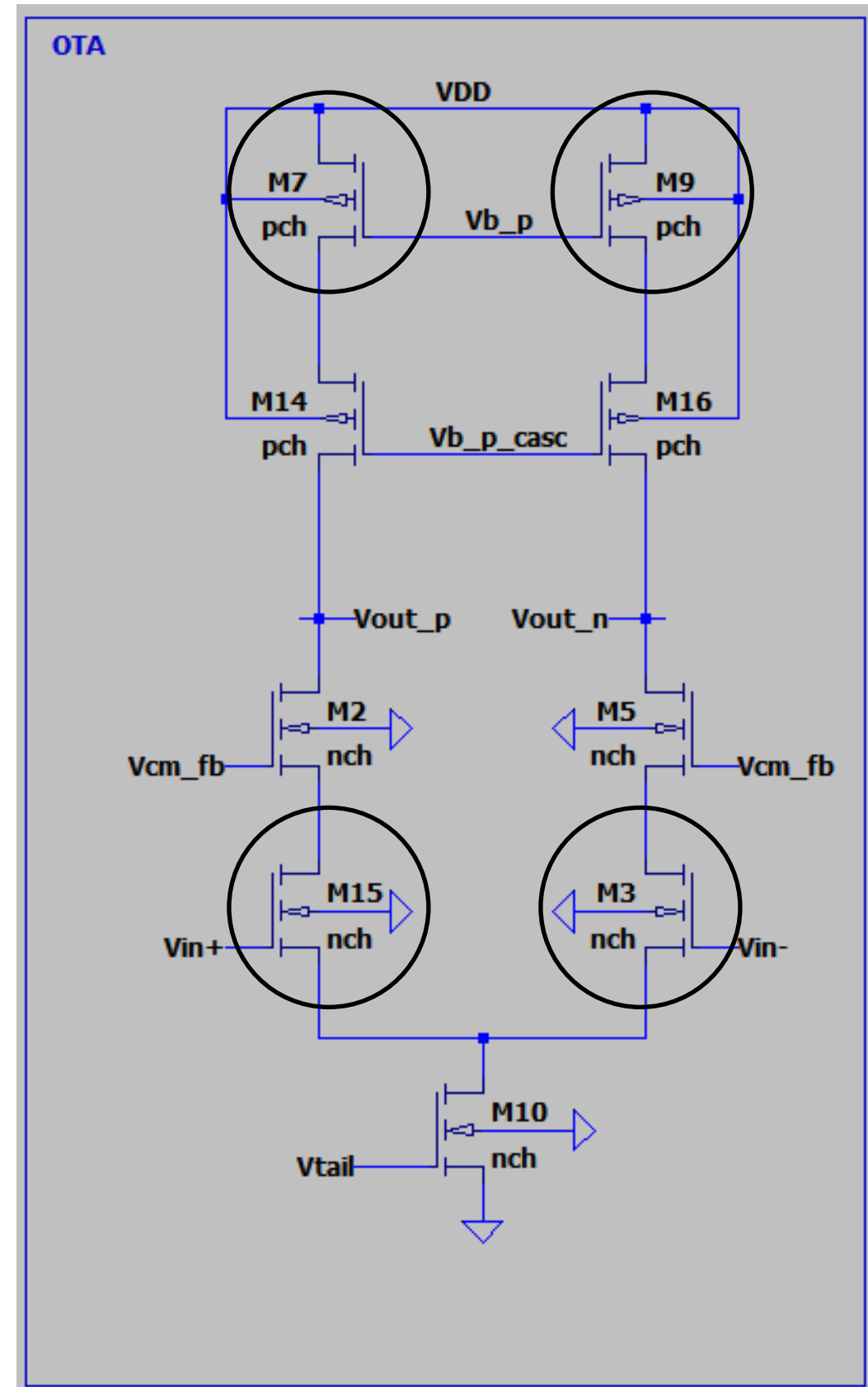
Use the file on Brightspace: **Week6_Ex2.asc**. It contains the setup to simulate the noise of a telescopic amplifier. Note that all sizes are parametrized to help scaling. The goal of this exercise is to identify dominating noise sources and see how we can change the design to improve the noise behaviour.

A. Find the dominating noise sources

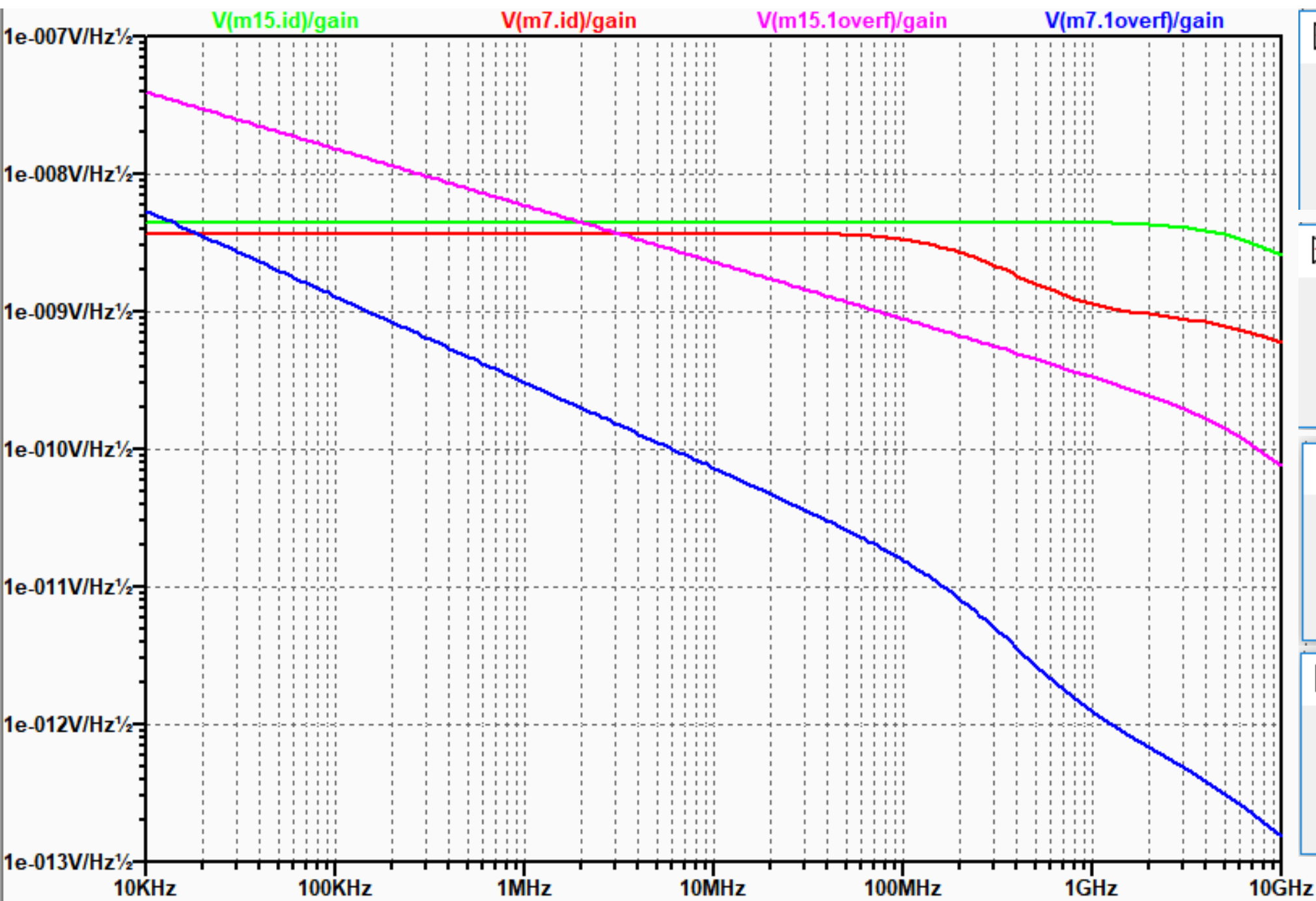
- Perform a noise simulation, the noise of which 4 transistors dominate?
- Show a plot of the input referred thermal noise and flicker noise voltage spectral density of these 4 transistors.
Hint: To plot the flicker noise or thermal noise only, right-click on the plot and look under 'add traces'.
 - $V(mX.1overf)$ is flicker noise
 - $V(mX.ds)$ is thermal noise
- Also give the integrated input referred RMS noise voltage of all 4 transistors.
Hint: The integrated noise can be found using 'cntrl+left-click' on the trace name in the plot window.

Results - Exercise 2.A

- **Dominating transistors**
M15 M3
M7 M9
- Contribution of cascode transistors is small.
- Bias network does not add to the noise.



Results - Exercise 2.A



$V(m15.id)/gain$	✕
Interval Start:	10KHz
Interval End:	10GHz
Total RMS noise:	367.09μV

$V(m7.id)/gain$	✕
Interval Start:	10KHz
Interval End:	10GHz
Total RMS noise:	100.51μV

$V(m15.1overf)/gain$	✕
Interval Start:	10KHz
Interval End:	10GHz
Total RMS noise:	28.761μV

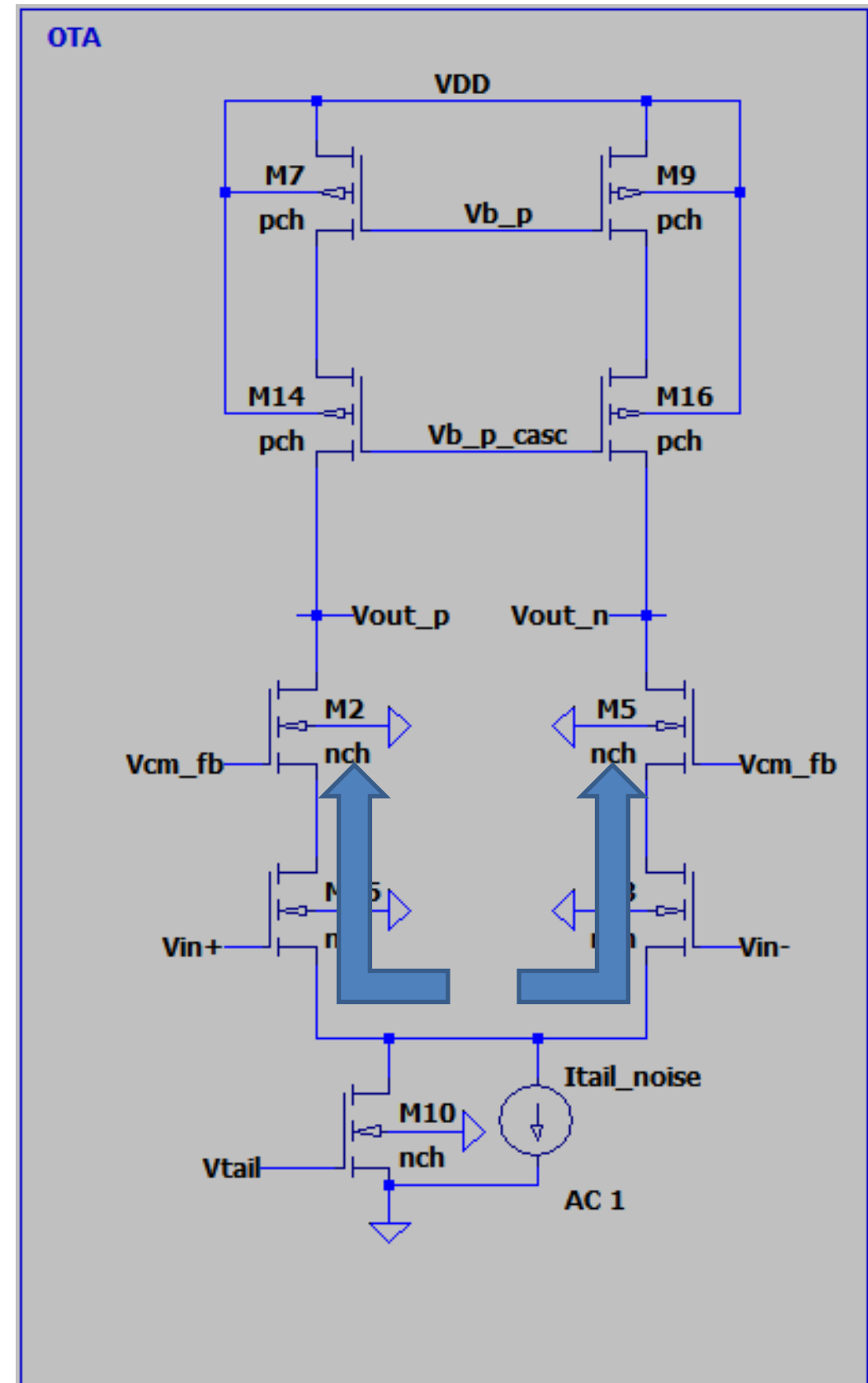
$V(m7.1overf)/gain$	✕
Interval Start:	10KHz
Interval End:	10GHz
Total RMS noise:	1.0444μV

Exercise 2.B – LTspice

- What is the noise contribution of the NMOS which provides the tail current (M10) and the noise contribution of the biasing network? Explain why.

Results - Exercise 2.B

- The noise from the tail transistor is split equally over the two branches.
 - Voltage on v_{out_p} and v_{out_n} caused by the noise from the tail current are equal.
- Noise only appears in common mode, not in differential mode**



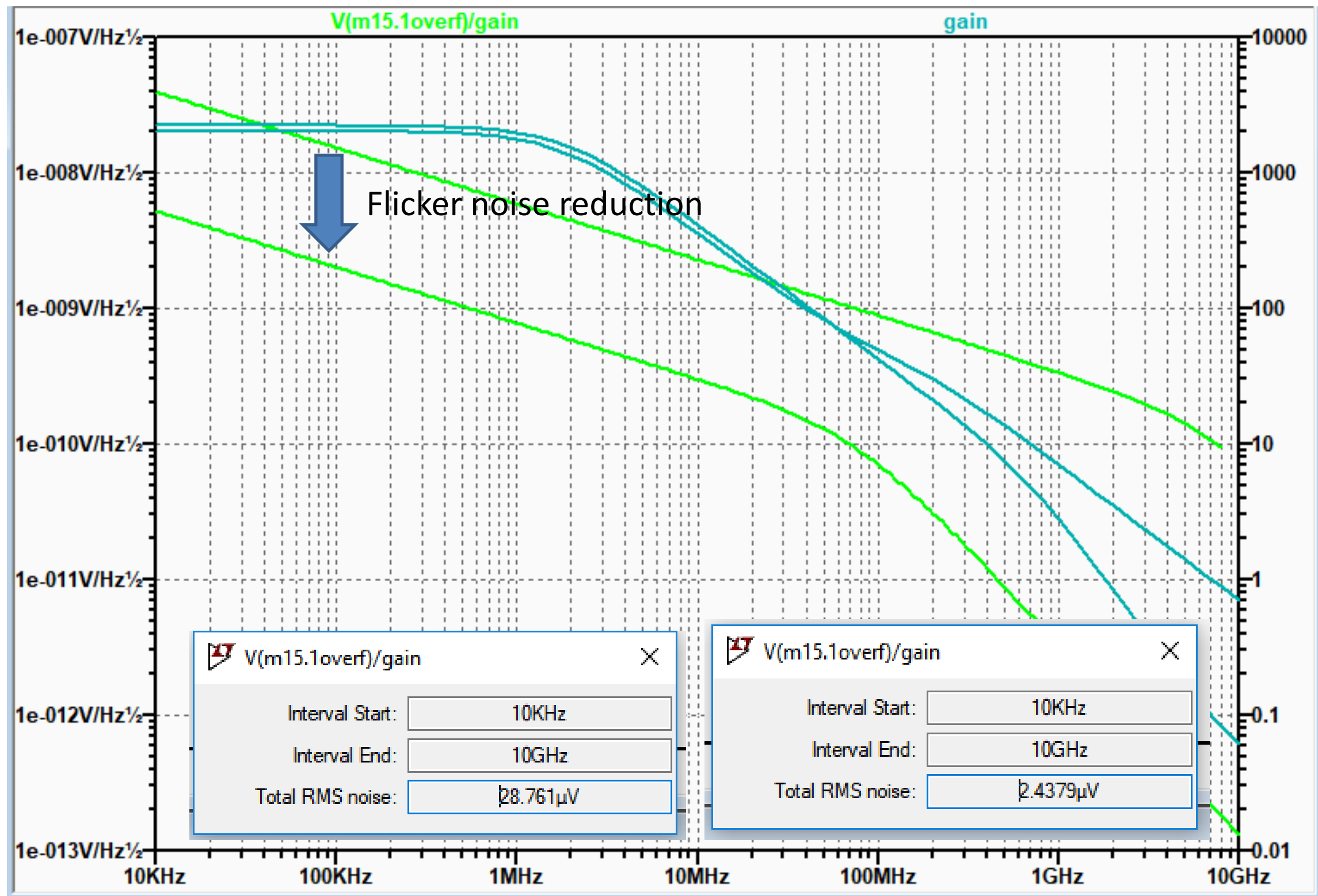
Exercise 2.C – LTspice

- For low frequencies, the Flicker noise dominates. We want to reduce the noise at these low frequencies.
- Scale the input transistors (M3 and M15) such that the input referred flicker noise becomes lower than $3 \mu V_{RMS}$, while maintaining a gain larger than 63 dB and without increasing the power consumption.
- *Hint: Use parameters at top of schematic*

Results - Exercise 2.C

- Recall the equation for flicker noise: $\overline{V_{n,1/f}^2} = \frac{K}{C_{ox}WL} \cdot \frac{1}{f}$
- By increasing both the width and the length of the transistor, the W/L -ratio does not change. The gain of the amplifier is not affected (first order approximation).
- We start at $28.76 \mu V_{RMS}$
- Required reduction: $\sim 10x$ in RMS value $\rightarrow \sim 100x$ in power
- W and L both increase with factor $10x$
- The gain is reduced slightly: from $\sim 2200x$ to $\sim 2100x$ for $9x$ increase and $\sim 2000x$ for $10x$ increase of W and L (g_m does decrease in practice).
- **Large input transistors are good for noise behavior.**
What is the cost when increasing transistor size?

Results - Exercise 2.C



Exercise 2.D – LTspice

- At this point, the thermal noise dominates over the flicker noise.
- Reduce the thermal noise of the input transistors by a factor 3. You are allowed to increase the power consumption, but keep the transistors in the same operating region.
Furthermore, the gain should not (significantly) be affected.

Results - Exercise 2.D

- To reduce the (thermal) noise, we increase the W of all transistors. The output impedance decreases, but the g_m increases. Use:

$$\overline{I_{n,therm}^2} = 4kT\gamma g_m [I^2/Hz]$$

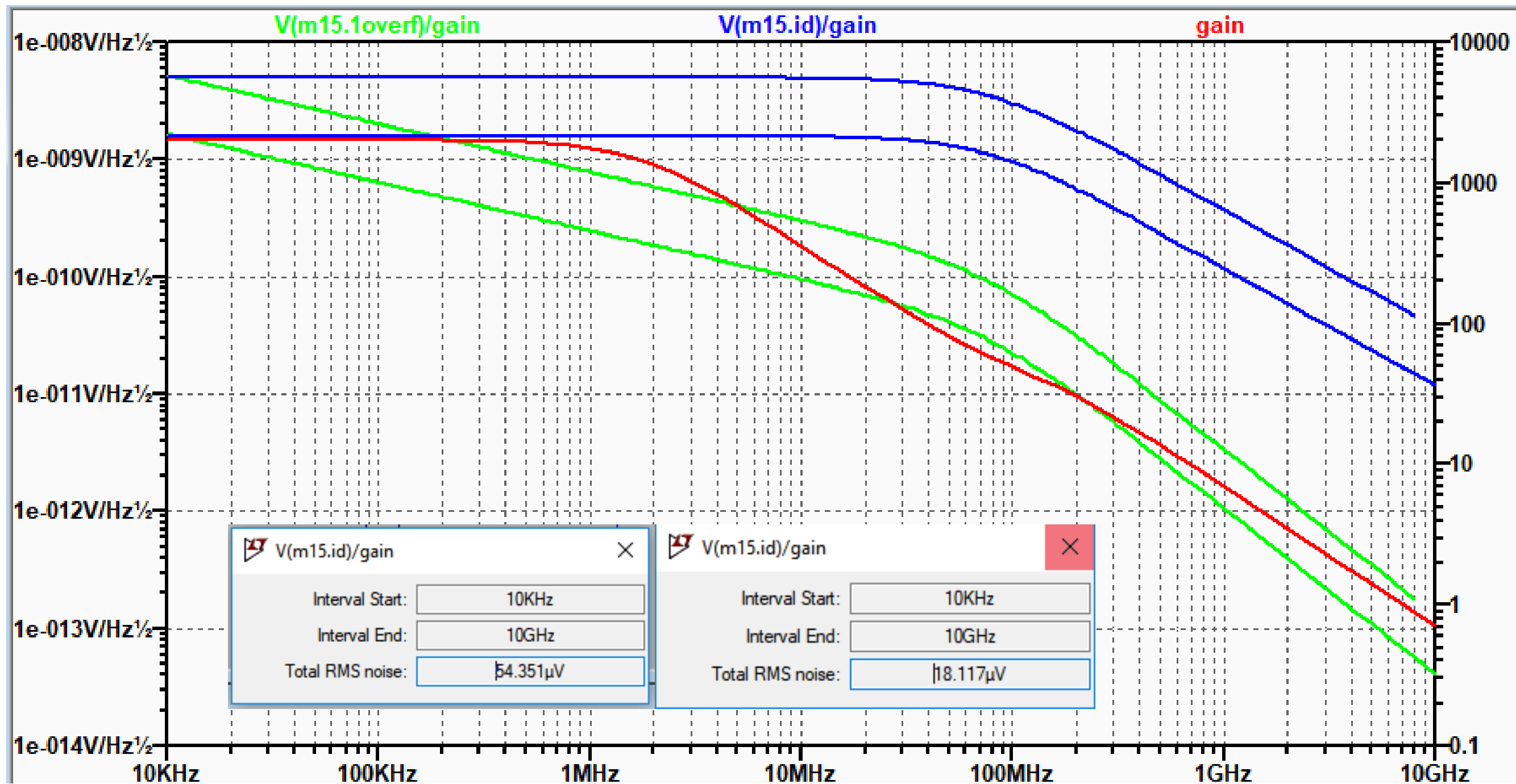
- When increasing the width of the transistor, the g_m , as well as the squared noise current, increase proportionally.
- The output impedance of the amplifier decreases proportionally.
- Using:

$$\overline{V_{n,therm}^2} = 4kT\gamma g_m \cdot r_{out}^2 [V^2/Hz]$$

we find that the output noise power spectral density decreases proportionally by increasing W.

- To decrease the RMS noise by a factor $3x$, we need to increase the width of all transistors by a factor $9x$.
- **Lesson: To get rid of thermal noise, the width of all transistors can be increased. This increases the power consumption.**

Results - Exercise 2.D

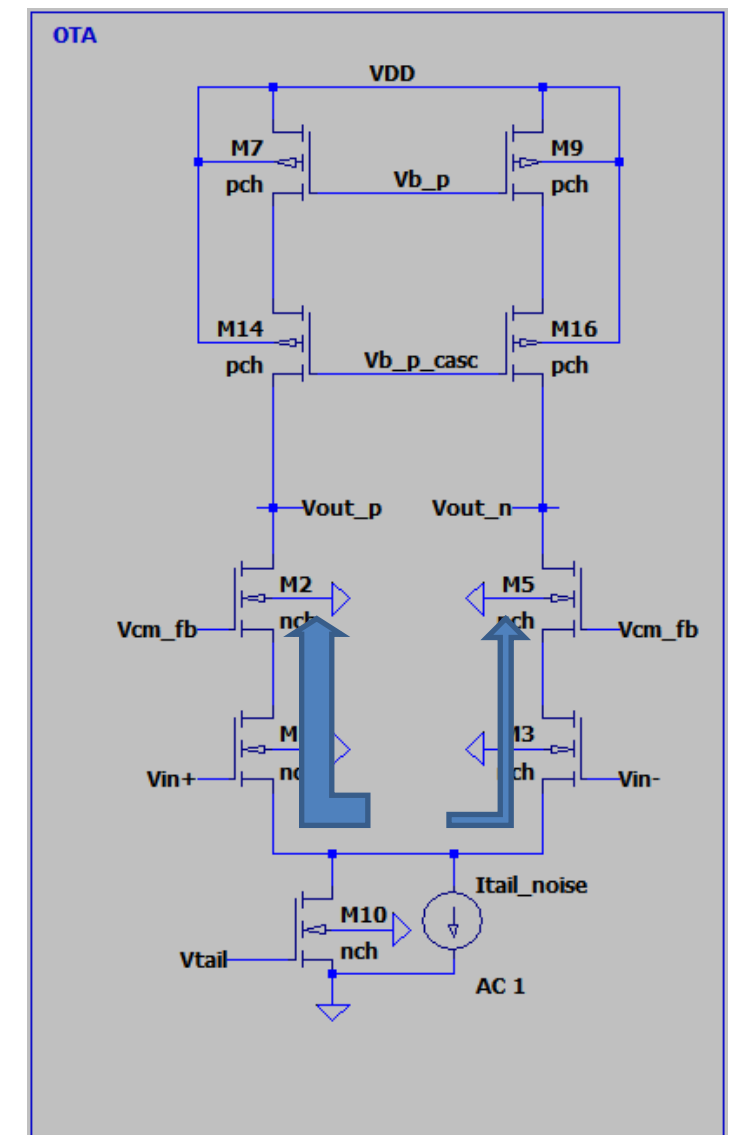
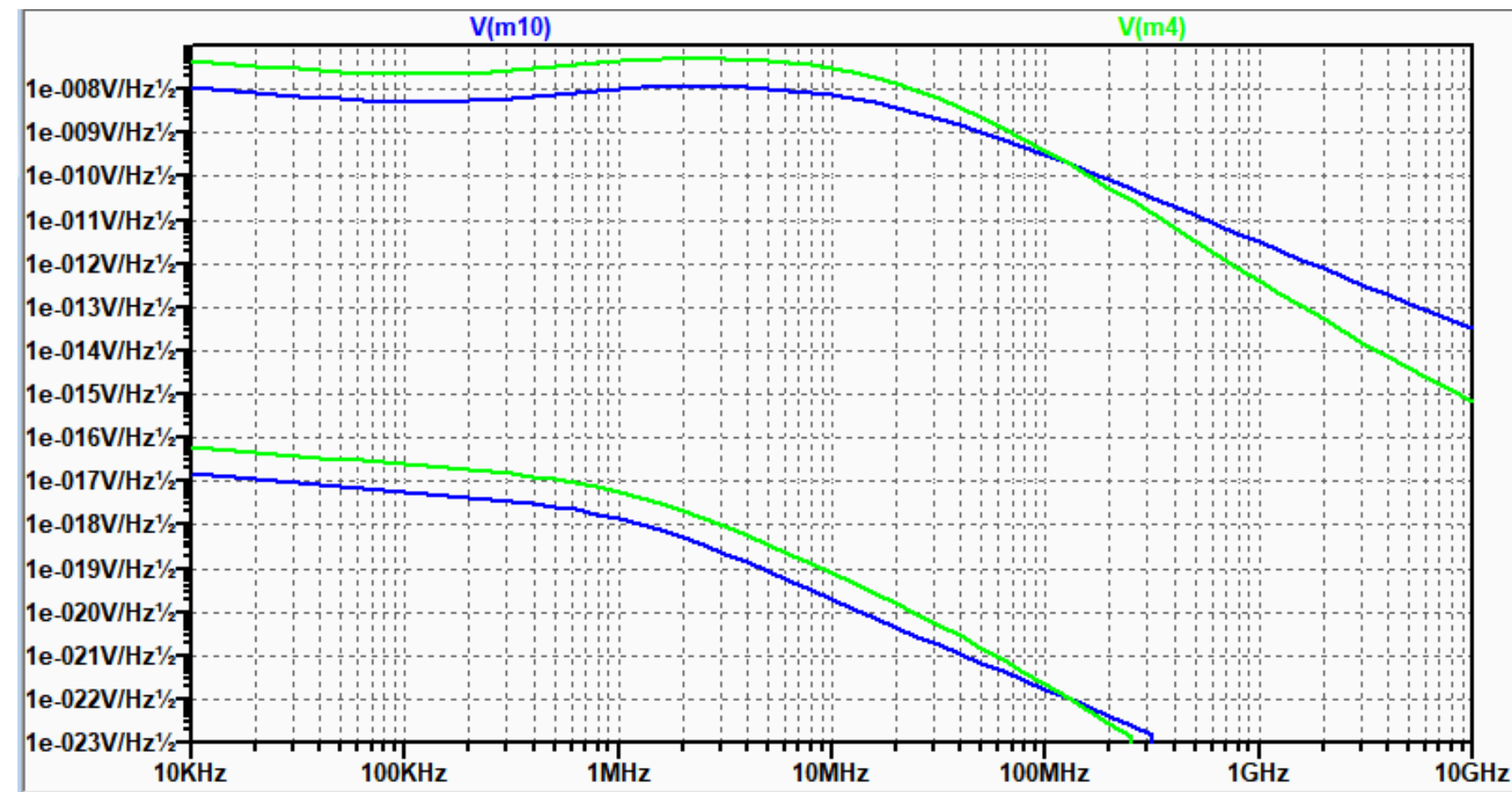


Exercise 2.E – LTspice

- Up until this point we used a perfectly symmetric amplifier. In reality, the amplifier will not be completely symmetric.
- Introduce a DC differential bias at the input of 20 mV. What happens to the noise from the tail current and the connected current mirror (M10 and M4)? Why?
- Explain the difference between the noise from M10 and M4.

Results - Exercise 2.E

- The noise current from the tail transistor does not split equally between the two branches.
- The common mode gain from M10 source to $V_{out,p}$ and $V_{out,n}$ is not equal anymore. The difference appears as differential mode noise.



Summary

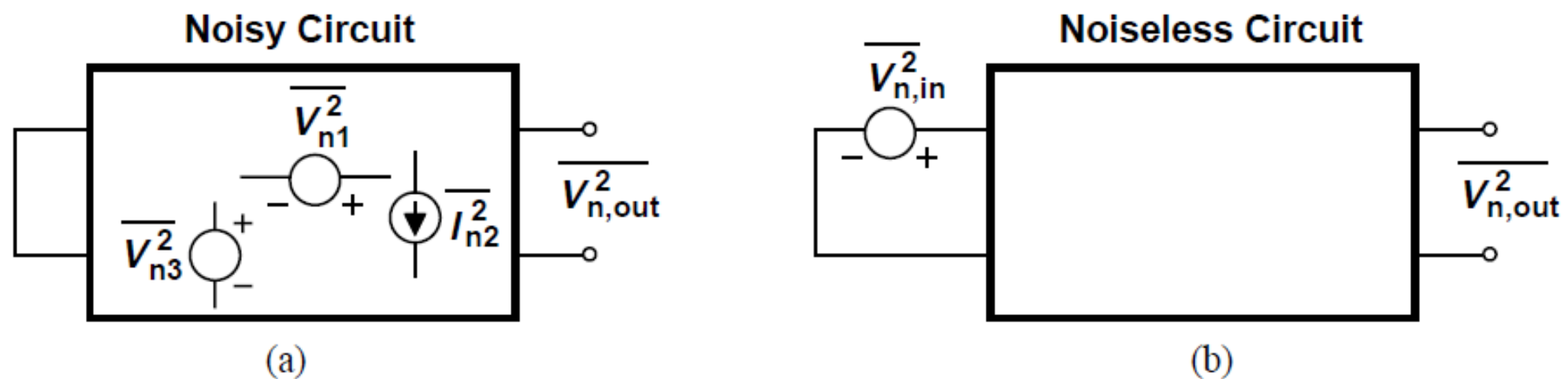
- This week
 - How to deal with random signals (average power, PSD,...)
 - Types of noise in CMOS circuits (thermal, flicker)
 - How to compute noise effects in circuits
- Spice simulators
 - Simulating noise
 - Simulate input/output referred noise
 - Analyze noise breakdown

Recipe for output noise calculation

- Steps:
 1. Disable all independent sources (V and I)
 2. Add the sources for noise
 3. Compute the transfer function $H_i(s)$ from each noise source to the output (as in any linear circuit)
 4. Compute the individual contribution as $S_i(f)|H_i(j2\pi f)|^2$
($S_i(f)$ is the PSD of noise source i)
 5. Sum up all the contributions

Input-referred noise

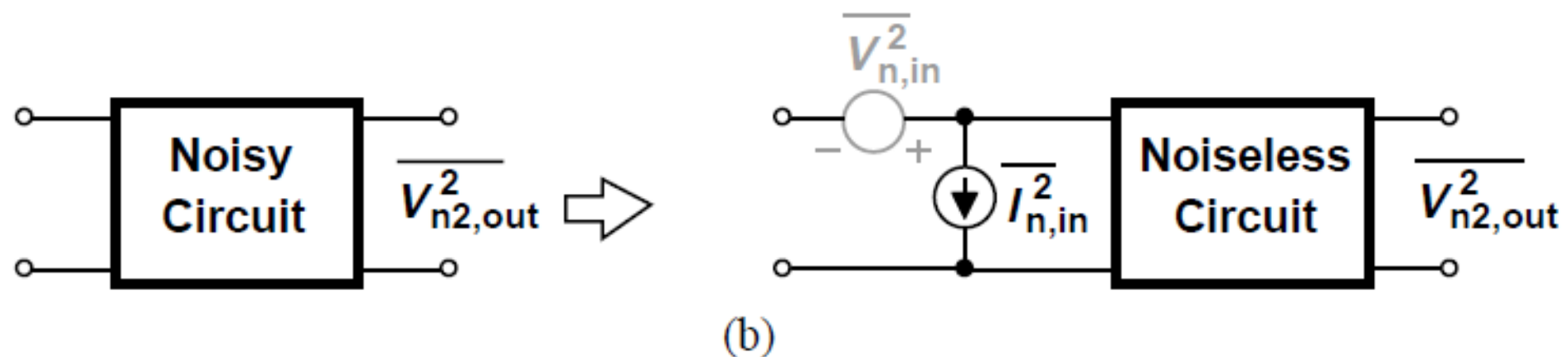
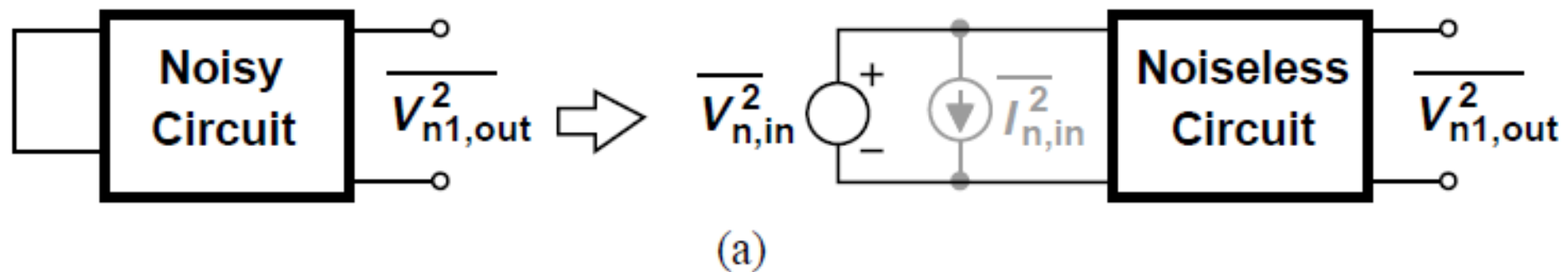
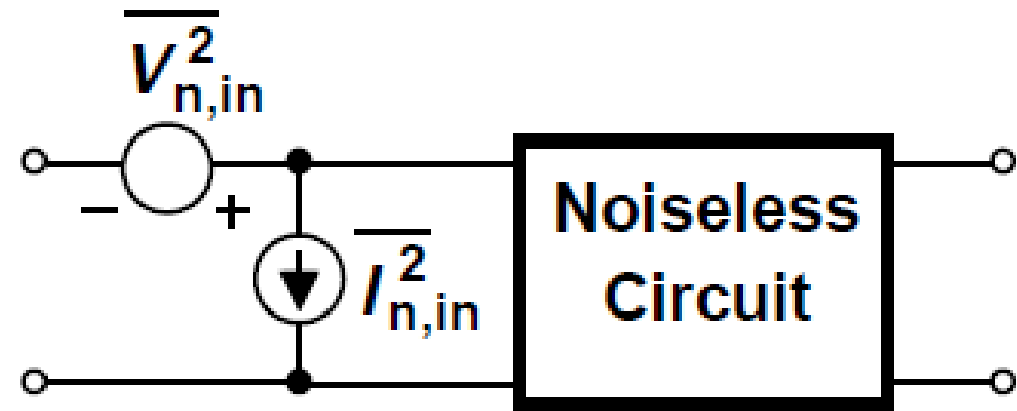
- Simple case (zero source impedance)



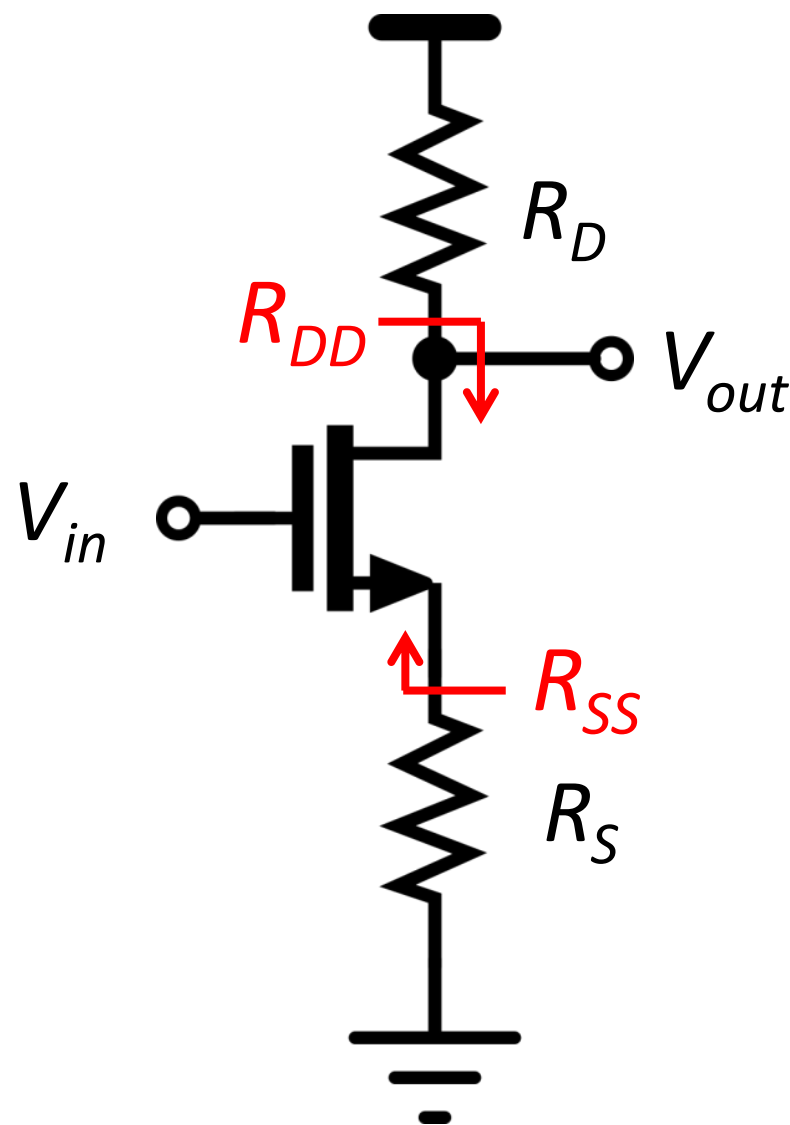
$$\overline{v_{n,out}^2} = A_v^2 \overline{v_{n,in}^2}$$

Input-referred noise

- General case



Useful relations



- Resistance from drain

$$R_{DD} = r_o[1 + (g_m + g_{mb})R_S] + R_S$$

$$\cong r_o + R_S + g_m r_o R_S$$

$$R_{SS} = \frac{r_o + R_D}{1 + g_m r_o} \cong \frac{1}{g_m}$$

- Norton-equivalent from drain (w/o R_D)

$$G_m = \frac{g_m r_o}{r_o[1 + (g_m + g_{mb})R_S] + R_S}$$

$$\cong \frac{g_m}{1 + g_m R_S}$$

- Approximations assume

- $r_o \gg R_D, R_S$
- Neglect g_{mb}